

From Timing-Limited to Optics-Limited: Synchronization-Aware Validation of Camera-Based Yaw-Rate Estimation on a Spacecraft Attitude Testbed

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Abstract—Ground testbeds for spacecraft attitude determination and control often rely on an overhead camera as a source of ground-truth motion, which assumes the camera is accurate over the full range of motion exercised during a test. This paper reports a synchronization-aware validation of camera-based yaw angular-rate estimation on a single-axis rotational testbed. A monocular camera observes AprilTags mounted on the rotating body; the per-tag yaw measurements are fused and differentiated to obtain angular rate, which is compared against two independent references: a 1496-count magnetic encoder logged at 100 Hz and a rate-output inertial measurement unit (IMU) on a commercial ADCS board (ESAT). Ten setpoints between 30 and 300 deg/s were each repeated three times, and errors are reported over the constant-speed portion of each run. We find that inter-sensor timing, rather than the optics, governed the measured error until it was handled explicitly. After software synchronization the camera agreed with the encoder to 0.87 deg/s at 30 deg/s and to about 2–3% of the commanded rate up to 300 deg/s, with correlation of 0.99 or higher and camera–encoder bias magnitude below 0.6 deg/s. The two references agreed with each other to within 1.3–2.0 deg/s with no systematic dependence on speed, which places the residual high-rate error in the camera stream rather than in the references. In a paired re-analysis of all 30 runs, estimating the alignment without the constant-speed segment changed the plateau RMS by at most 0.031 deg/s. Exposure was held fixed at 5 ms, so motion blur and residual frame-timestamp jitter both remain consistent with the observed growth and are not separated by this dataset.

Keywords—vision-based angular rate, AprilTag, fiducial markers, attitude determination, ground testbed, sensor synchronization, encoder, IMU, validation

I. INTRODUCTION

Air-bearing platforms and motor-driven rotational rigs are commonly used to develop and validate attitude determination and control systems (ADCS) for small spacecraft. Many such facilities use a camera as the trusted external reference, so that the vision system supplies the motion against which onboard sensors and control laws are judged [6], [7]. This is convenient, but it quietly assumes that the camera is accurate enough for that role, and across the full range of motion the testbed will exercise.

This paper examines that assumption for yaw angular rate and answers it with measurements rather than specifications. Instead of treating the camera as ground truth, we treat it as the device under test and ask how accurately a low-cost AprilTag-based system estimates yaw rate, how that accuracy changes as the rotation speeds up, and what limits it.

The study began with a discouraging result: early comparisons showed rate errors of roughly 5–8 deg/s. On inspection almost none of that error was optical. The reference had been logged at a low rate with a coarsely quantized velocity field, and the two streams were not placed on a common time base. Once the reference was logged at 100 Hz with timestamps and the streams were aligned, the apparent error fell by roughly a factor of five. That experience shaped the rest of the work, and synchronization is treated here as part of the validation method rather than an implementation detail.

The contribution is therefore not the use of AprilTags for pose estimation, which is well established [1], [2], nor a new estimator. It is threefold: a synchronization-aware procedure for validating vision-based rate estimation against independent references; a quantified accuracy envelope for yaw rate from 30 to 300 deg/s, with repeats; and the finding, supported by a third independent sensor, that the residual error at high rate is camera-side. The

contrast in the title refers to where the measured error was found to reside, first in the inter-sensor comparison and afterwards inside the camera stream. It is not a claim that optical effects have been separated from the remaining camera-side timing effects, which Section VII explains the present dataset cannot do. We restrict the scope to yaw rate. Yaw here is rotation about the camera boresight, which for a fronto-parallel planar tag is a well-conditioned axis, whereas pitch and roll are harder, as discussed in Section VIII.

II. RELATED WORK

This study is positioned within four relevant areas of prior work. The first is fiducial-marker pose estimation. AprilTag [1], [2] and ArUco [3] are the standard square-marker systems, and several studies compare their pose accuracy under controlled conditions, including simulated motion blur [4]. Detailed analyses of AprilTag state estimation report that the apparent yaw of a tag, together with oblique and distant viewing, is among the largest sources of static pose error [5]. That analysis concerns a camera yawing relative to a tag on the ground, which is a different geometry from ours: on the present rig the tag rotates in the image plane about the camera boresight and stays nearly fronto-parallel. We nonetheless take from it the motivation to combine several tags per frame, so that no single noisy detection dominates, and we characterize the resulting rate error against an external reference rather than the static pose error. The blur results in [4] are obtained by applying blur synthetically to otherwise static imagery, so they establish that the mechanism degrades marker pose without providing an angular-rate error against a moving reference.

Dynamic use of AprilTags is less well characterized than static use, and the gap is one of reference and timing rather than of detection. Maragh used AprilTags to control the position and angular velocity of a rotating body and to find the highest velocity

at which a low-cost camera could still detect the tags [14]; there the tag measurement is itself the feedback signal, so no independent rate reference is involved and no timing treatment is required. Nissler et al. evaluated multi-tag pose against a laser tracker and restricted the study to static poses because synchronization between the tracker and the vision system was not available to them [15], which is the same obstacle addressed here. Closest to the present work, Liss et al. built an AprilTag attitude truth system for a three-degree-of-freedom air-bearing CubeSat testbed and added an extended Kalman filter to suppress the large attitude excursions the raw detections show while the platform is moving [13]. They report converged filter standard deviations of about 1 deg in the Euler angles and 2.5 to 3.5 deg/s in the body rates, at yaw rates of about 30 deg/s and below. Those rate figures are filter standard deviations rather than errors against an independent rate sensor, and the timing relationship between the

vision system and any other measurement is not part of the evaluation.

The same limitation recurs across this literature. Fiducial-marker accuracy for attitude has been characterized in detail under static or quasi-static conditions, and marker-derived rates have been reported as filter outputs or used as control feedback, but we are not aware of a study that reports marker-derived angular-rate error against an independently timed hardware rate reference across a decade of rate. That comparison is what this experiment adds: a rate envelope from 30 to 300 deg/s with three repeats per setpoint, measured against a 100 Hz encoder and cross-checked against a second independent sensor, with the inter-sensor time alignment treated as part of the method and its effect on the plateau RMS measured rather than assumed. At 30 deg/s, the top of the range covered in [13], the camera here agrees with the encoder to 0.87 deg/s. Table I summarizes the comparison.

TABLE I. DYNAMIC VALIDATION OF FIDUCIAL-MARKER ATTITUDE AND RATE ESTIMATION: SCOPE OF PRIOR WORK

Study	Method	Dynamic scope	Reference / timing	Reported performance / gap
Maragh, 2013 [14]	AprilTag; rotating body	Detection-limit study	No external reference	Closed-loop demonstration; no calibrated rate error
Nissler et al., 2016 [15]	Multiple AprilTags	Static only	Laser tracker; synchronization unavailable	Static pose only; identifies timing barrier
Kalaitzakis et al., 2021 [4]	AprilTag/ArUco/ARTag/STag	Synthetic blur on static imagery	Controlled static poses	Blur mechanism characterized; no rate reference
Liss et al., 2018 [13]	AprilTag + EKF; 3-DOF air bearing	Yaw up to about 30 deg/s	No independent rate reference	Body-rate SD 2.47–3.49 deg/s; not external-reference error
Modenini et al., 2020 [7]	Metrology vision system	Air-bearing verification	Vision system is reference	Attitude, not rate
Ornati et al., 2025 [6]	Active LED markers	Static-accuracy emphasis	High-fidelity optical simulation	Arc-second attitude accuracy; no independent rate reference
This work	Printed tag36h11; seven fused/frame	30–300 deg/s; 10 setpoints $\times$ 3	1496-count encoder @100 Hz + ESAT gyro	Camera–encoder RMS 0.87 – 6.64 deg/s; alignment effect $\leq$ 0.031 deg/s

The second line is vision-based attitude estimation for spacecraft simulators. The closest work estimates the attitude of an air-bearing platform from a monocular camera and sets of fiducial LED markers, using a geometry-based iterative algorithm and reaching arc-second-level accuracy, with performance assessed against high-fidelity simulations of the same optical system [6]. That study targets high static accuracy with bespoke active markers; the present work targets the dynamic rate envelope with low-cost printed tags and foregrounds synchronization against independent hardware references. The two are complementary. The third line is ADCS testbed development. Air-bearing and rotational testbeds for nanosatellites have a long history [8], and modern facilities commonly integrate a metrology vision system as the ground-truth source [7]. A recent review of CubeSat ADCS testbeds places the present UAEU rig within this broader verification program [12]; in those facilities the camera is trusted, whereas here it is the subject of the test. The fourth line is multi-sensor validation practice: comparing a sensor against one or more independent references and reporting agreement metrics. Our contribution within that line is the explicit treatment of inter-sensor timing and the use of a second independent reference to decide whether residual error belongs to the device under test or to the references. The paper does not claim a new marker, detector, or estimator; it claims a validation methodology and an empirical characterization that the adjacent literature does not provide for the rate domain, together with an explicit account of what that methodology can and cannot establish about the source of the residual error.

III. EXPERIMENTAL SETUP

The experimental platform is a single-axis motor-driven rotational rig designed specifically to evaluate yaw-rate measurements; it is not intended to reproduce the full three-axis dynamics of an air-bearing spacecraft simulator. A DC gear motor drives a horizontal

platform through a BTS7960 H-bridge. A magnetic quadrature encoder on the output shaft provides 1496 counts per revolution after decoding. An Arduino microcontroller performs the decoding, runs a feedforward-plus-PI speed controller, and logs the encoder state. A monochrome Basler ace acA1920-40um camera (1920 $\times$ 1200 pixels) with an 8 mm lens views the platform from a fixed overhead mount (Fig. 1). The sensor uses a global shutter, so rolling-shutter skew is not a factor in the results below. The camera runs at about 33 frames per second in Mono8 with manual exposure (5 ms) and gain, and with auto-exposure and auto-gain disabled so that frame timing and image statistics remain constant during a run.



Fig. 1. Testbed overview. The Basler camera is mounted overhead on the aluminum-extrusion frame and views the tagged platform on the rotation stage below.

Seven AprilTags of the tag36h11 family (70 mm) are mounted on the rotating platform around a central marker (Fig. 2), so that several tags are visible at once and their yaw estimates can be fused. The second independent rate reference is the gyroscope-derived rotational-speed output of the attitude determination and control subsystem of the Theia Space Educational Satellite (ESAT), a commercial board carrying a six-axis inertial measurement unit and a three-axis magnetometer. Rotational speed is computed from the gyroscope measurements, and the manufacturer specifies a rotational-speed accuracy of 1 deg/s [17]. Its approximately 1 Hz update rate reflects the telemetry logging path used in this experiment rather than the gyroscope bandwidth. Ten rate setpoints were commanded, 30 to 300 deg/s in 30 deg/s steps, each run three times for thirty runs. Every run follows the same profile: a short rest, a ramp to the commanded rate, a constant-speed hold, and a return to rest.



Fig. 2. Top-down view of the rotating platform carrying seven tag36h11 markers (70 mm). The central and six peripheral tags are fused per frame to reduce yaw jitter.

IV. CALIBRATION AND SYNCHRONIZATION

A. Camera calibration

Intrinsic calibration used a printed checkerboard (9×6 inner corners, 20 mm squares) and 39 valid views of 40, following the standard planar-target method [9]. The recovered parameters are listed in Table II, and the root-mean-square reprojection error was 0.16 px. Each frame is undistorted with these parameters before tag detection, and the same intrinsics are passed to the pose solver.

TABLE II. SYSTEM CALIBRATION AND ACQUISITION SETTINGS

Parameter	Value
Focal length f_x, f_y (px)	1443.25, 1443.74
Principal point c_x, c_y (px)	963.22, 580.33
Distortion (k_1, k_2, p_1, p_2, k_3)	-0.158, 0.162, -0.0005, -0.0002, -0.118
RMS reprojection error (px)	0.16
Calibration views	39 / 40
Tag family / size	tag36h11 / 70 mm
Frame rate / exposure	~33 fps / 5 ms
Encoder resolution	1496 counts/rev
Encoder log rate	100 Hz
ESAT output / rate	gyroscope-derived angular rate / ~1 Hz

The encoder angle used as the rate reference follows directly from the decoded count,

$$\theta(t) = (2\pi/N) n(t) \quad (1)$$

where θ is the platform angle, n is the decoded count, and $N = 1496$ counts per revolution. The encoder angle is differentiated offline using the same Savitzky-Golay operator applied to the camera yaw in Section V.

B. Synchronization

The camera, encoder, and ESAT rate stream are not hardware-synchronized, so the three streams must be placed on a common time base before any error is computed. Each camera frame is timestamped on the host immediately after it is retrieved from the acquisition library, not by the camera itself, so the interval between exposure and timestamp includes sensor readout, transfer, and host scheduling and is not guaranteed to be constant. The encoder timestamp is taken on the microcontroller immediately after an atomic read of the decoded count. The ESAT rotational-speed output is logged through a separate telemetry path at about 1 Hz; the origin of its recorded timestamp is not established by the archived data.

First, the encoder is logged with microsecond timestamps at 100 Hz, and its angular rate is reconstructed offline from raw counts rather than from any onboard rate estimate, which avoids the coarse velocity quantization produced by a short-window onboard calculation. Second, because each run contains a single clean rest-to-motion-to-rest profile, the camera stream is aligned to the encoder by matching the onset of motion in both rate signals and then refining the offset by minimizing their squared difference over the rising edge, plateau, and falling edge,

$$\Delta t^* = \arg \min_{\Delta t} \sum_k [\hat{\omega}(tk + \Delta t) - \omega_{enc}(tk)]^2 \quad (2)$$

In (2), Δt^* is the estimated camera-to-encoder time offset, Δt is a trial offset, $\hat{\omega}(t)$ is the camera-derived yaw rate, $\omega_{enc}(t)$ is the encoder-derived yaw rate, and tk is the k th encoder timestamp. The ESAT stream is aligned to the encoder in the same way. Estimating an inter-sensor time offset by minimizing disagreement is standard practice in multi-sensor calibration [16].

Δt^* absorbs any constant delay in the camera path. One such contribution is the acquisition-stage low-pass filter of Section V: for $\alpha = 0.20$ at about 33 frames per second, the low-frequency group delay implied by the filter parameters is $(1 - \alpha)/\alpha \approx 4$ frames, or roughly 0.12 s. That figure follows from the filter coefficients and the frame rate; the delay between exposure and timestamp was not measured. Δt^* is therefore an effective alignment parameter for the rate comparison rather than an estimate of the physical clock offset between the devices, and we make no angle-domain claim from it.

The refinement window covers the rising edge, plateau, and falling edge, while the reported errors are computed on the trimmed plateau, so the two windows overlap. To test whether that overlap flatters the camera, all thirty runs were reprocessed under two alternative alignments through one identical processing chain and compared run by run: the onset offset with no refinement at all, and a refinement estimated from the rising and falling edges only, with the plateau excluded from the fit and held out for evaluation. The three offsets agree closely, the refined value differing from the onset value by at most 83 ms and from the transition-only value by at most 20 ms. Because the comparison is paired within each run and uses the same reconstructed processing chain throughout, only the differences between alignment strategies are

reported from this re-analysis; Tables IV and V retain the accepted absolute results.

Excluding the plateau from the fit changes the plateau RMS by +0.005 deg/s on average across the thirty runs and by at most 0.031 deg/s in any single run. That largest absolute change occurs at 300 deg/s and is 0.65% of the RMS there. The largest relative change is 3.07%, at 30 deg/s, where the corresponding absolute change is 0.023 deg/s. Onset-only alignment, which involves no least-squares fit, changes the plateau RMS by -0.018 deg/s on

average; refinement lowers the plateau RMS in 9 of the 30 runs and raises it in 21, and the offset that minimizes squared disagreement over the full motion profile does not generally coincide with the offset that would minimize the plateau RMS. Perturbing the offset by one frame changes the plateau RMS by 0.50% on average, with the largest sensitivity at the lowest rates, where the absolute errors are smallest. The plateau accuracy reported in Section VI is therefore not produced by fitting the offset. Table III summarizes the comparison.

TABLE III. EFFECT OF ALIGNMENT STRATEGY ON THE REPORTED PLATEAU ERROR (PAIRED OVER THIRTY RUNS)

Alignment strategy	Plateau in fit?	Mean Δ RMS (deg/s)	Max $ \Delta$ RMS (deg/s)	Max $ \Delta$ RMS / RMSref (%)	Mean Δ bias (deg/s)
Full-profile refinement (reported)	Yes	—	—	—	—
Onset only, no refinement	No	-0.018	0.145	4.19	+0.007
Transitions only; plateau held out	No	+0.005	0.031	3.07	-0.001

Note: the maximum absolute and maximum relative changes occur in different runs. For transition-only alignment, 0.031 deg/s occurs at 300 deg/s (0.65% of that run's RMS), whereas 3.07% occurs at 30 deg/s and corresponds to 0.023 deg/s.

We verified on every run that the refinement offset was small and that the aligned rate signals correlated at 0.99 or higher. Because Δt^* is chosen to maximize agreement over the same motion segment, this correlation is an agreement metric: it indicates that the alignment converged and that the two profiles share a common shape, not that the streams are correctly timed in an absolute sense. That the reported plateau accuracy is insensitive to the alignment procedure is established by the re-analysis above. Separately, the encoder-ESAT comparison of Section VI provides an independent check that the reference disagreement does not grow with speed. The importance of synchronization is nevertheless substantial: with the encoder logged at a low rate and the streams unaligned, the same data produced rate errors several times larger than after synchronization. The early large camera error was, to first order, a timing artifact.

V. YAW-RATE ESTIMATION

For each detected tag, the pose rotation matrix is converted to a yaw angle,

$$\psi(i) = \text{atan2}(R_{21}(i)(t), R_{11}(i)(t)) \quad (3)$$

where $R(i)$ is the rotation matrix of tag i . Each tag yaw is unwrapped independently and referenced to its value at first detection, so that all tags report motion relative to a common start. The per-tag relative yaws are fused into a single estimate by a decision-margin-weighted average over the set $V(t)$ of tags that agree with the median within the threshold $\tau = 12^\circ$,

$$\hat{\psi}(t) = [\sum_{i \in V(t)} w_i \tilde{\psi}(t)] / [\sum_{i \in V(t)} w_i] \quad (4)$$

where w_i is the detection decision margin of tag i . Fusing several tags reduces the per-frame jitter relative to any single tag and removes the discontinuities that occur when individual tags drop in and out of view. During acquisition, the fused camera yaw was passed through a first-order infinite-impulse-response (IIR) low-pass filter,

$$y_k = \alpha x_k + (1 - \alpha)y_{k-1}, \quad \alpha = 0.20 \quad (5)$$

where x_k is the current fused yaw, y_k is the filtered output, and y_{k-1} is the previous filtered output. The filtered fused yaw is then differentiated to angular rate with a second-order Savitzky-Golay filter using a window of about 0.3 s on a common time grid [10],

$$\hat{\omega}(tk) = (1/\Delta t) \sum_j = -m \dots m c_j \hat{\psi}(tk+j) \quad (6)$$

where c_j are the Savitzky-Golay filter coefficients and Δt is the grid spacing. The same Savitzky-Golay differentiator is applied to

the filtered camera yaw and the encoder angle of (1). Thus, the derivative operator is matched for both signals, while the camera stream additionally contains the acquisition-stage IIR smoothing. Detection noise is present before the IIR filter and is attenuated by it, whereas timing error shifts samples along the common time grid. Detection and pose estimation use the OpenCV and AprilTag libraries [2], [11].

Three sensor pairs are compared: camera-encoder, camera-ESAT, and encoder-ESAT. For each run we report the root-mean-square error,

$$eRMS = \sqrt{[(1/K) \sum_k (\omega_{cam,k} - \omega_{ref,k})^2]} \quad (7)$$

the mean absolute error and the bias,

$$eMAE = (1/K) \sum_k |\omega_{cam,k} - \omega_{ref,k}| \quad (8)$$

$$b = (1/K) \sum_k (\omega_{cam,k} - \omega_{ref,k}) \quad (9)$$

and the correlation. Errors are computed only over the constant-speed window, with the first and last 1.5 s of the plateau trimmed to exclude transients; correlation is computed over the full motion segment, since a flat plateau carries little shape information. For the ESAT pairs, samples are restricted to the plateau band, and given the approximately 1 Hz telemetry rate, RMS, MAE, and bias are the informative metrics. To separate camera-side variation from comparison artifacts, we also form an encoder-independent rate-variation diagnostic as the standard deviation of the difference between a light and a heavy derivative of the fused yaw. Here, K is the number of samples in the selected evaluation window. Error signs are defined as Camera-Encoder = Camera - Encoder, Encoder-ESAT = Encoder - ESAT, and Camera-ESAT = Camera - ESAT. Because the ESAT comparisons use sparse, independently aligned samples, the three pairwise bias values are not expected to be exactly additive.

$$\sigma_{cam} = \text{std}(\hat{\omega}_{light}(t) - \hat{\omega}_{heavy}(t)) \quad (10)$$

VI. RESULTS

Across all ten setpoints and three repeats, thirty runs were processed and none were excluded. Table IV reports the camera-encoder agreement. The camera tracks the encoder closely: at 30 deg/s the agreement is better than 1 deg/s, rising to about 6.6 deg/s at 300 deg/s. Expressed as a fraction of the commanded rate, the error remains within an approximately 2–3% range across the tested setpoints. The correlation is 0.99 or higher at every speed. The camera-encoder bias remains small and negative across the

envelope, with magnitude below 0.6 deg/s and approximately 0.2% of the commanded rate, which is consistent with a small scale mismatch rather than a distinct high-speed additive bias.

TABLE IV. CAMERA VERSUS ENCODER YAW-RATE AGREEMENT (MEAN $\pm$ SD OVER THREE REPEATS)

Setpoint (deg/s)	RMS (deg/s)	MAE (deg/s)	Bias (deg/s)	Correlation
30	0.87 ± 0.05	0.68 ± 0.04	-0.06	0.991
60	1.48 ± 0.14	1.08 ± 0.11	-0.11	0.994
90	1.69 ± 0.38	1.22 ± 0.25	-0.17	0.995
120	3.05 ± 0.53	2.20 ± 0.38	-0.20	0.993
150	3.34 ± 0.69	2.38 ± 0.31	-0.27	0.993
180	3.93 ± 1.02	2.87 ± 0.77	-0.43	0.993
210	4.98 ± 1.15	3.39 ± 0.93	-0.45	0.992
240	4.83 ± 0.68	3.27 ± 0.37	-0.43	0.993
270	6.66 ± 1.32	4.60 ± 0.92	-0.54	0.990
300	6.64 ± 1.89	4.57 ± 0.84	-0.53	0.990

The two references agree with each other with no systematic dependence on speed: the encoder-ESAT RMS remains between 1.3 and 2.0 deg/s from 30 to 300 deg/s, of the same order as the 1 deg/s rotational-speed accuracy specified for the ESAT gyroscope [17]. The approximately 1 Hz telemetry logging rate limits the ESAT stream to a supporting reference rather than a high-

resolution rate reference. The camera-ESAT RMS follows the camera-encoder curve and grows with speed. Table V lists both reference comparisons; correlation is omitted there because the sparse ESAT sampling makes it uninformative. Fig. 3 plots the RMS of all three pairs against speed.

TABLE V. REFERENCE COMPARISONS: ENCODER-ESAT AND CAMERA-ESAT (DEG/S, MEAN $\pm$ SD)

Setpoint	Enc-ESAT RMS	MAE	Bias	Cam-ESAT RMS	MAE	Bias
30	1.49 ± 0.06	1.20 ± 0.04	-0.56	1.30 ± 0.21	1.03 ± 0.14	-0.31
60	1.26 ± 0.39	0.96 ± 0.42	-0.43	1.79 ± 0.11	1.44 ± 0.10	-0.37
90	1.39 ± 0.02	1.13 ± 0.05	-0.23	2.21 ± 0.41	1.73 ± 0.40	+0.09
120	1.60 ± 0.10	1.34 ± 0.08	-0.24	3.11 ± 0.55	2.43 ± 0.44	-0.87
150	1.83 ± 0.28	1.48 ± 0.19	-0.54	2.97 ± 0.69	2.33 ± 0.51	-0.53
180	1.79 ± 0.17	1.49 ± 0.20	-0.10	4.64 ± 1.23	3.37 ± 0.74	-0.03
210	1.69 ± 0.11	1.43 ± 0.12	-0.32	6.17 ± 1.39	4.19 ± 0.94	+0.01
240	1.71 ± 0.06	1.43 ± 0.06	-0.17	5.95 ± 0.79	3.75 ± 0.19	+0.88
270	2.00 ± 0.40	1.71 ± 0.36	-0.28	6.33 ± 1.68	4.58 ± 1.41	+1.89
300	1.46 ± 0.17	1.20 ± 0.16	+0.03	5.41 ± 1.30	3.76 ± 0.30	+1.00

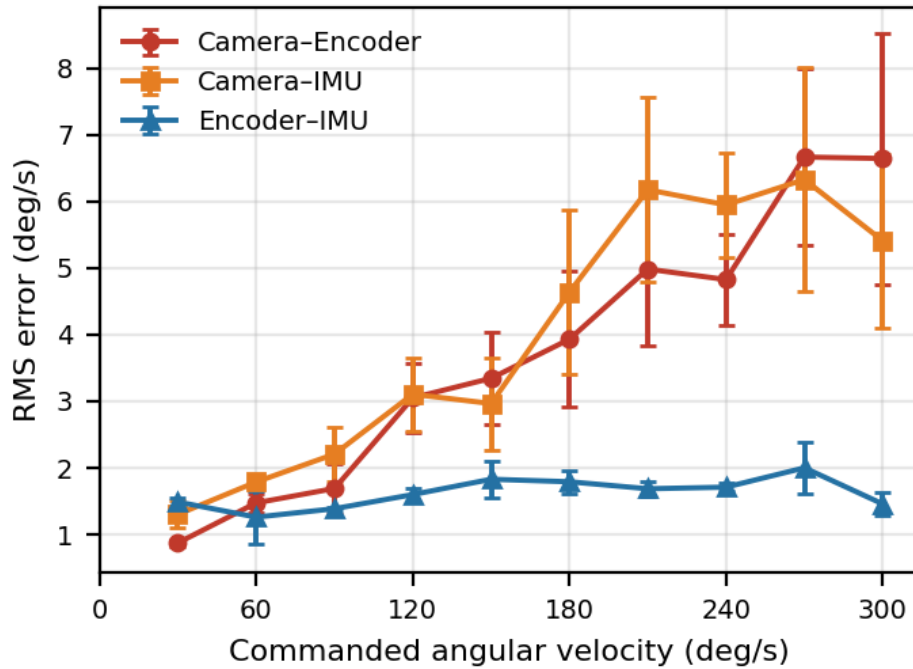


Fig. 3. Angular-rate RMS error versus commanded angular velocity for all three sensor pairs (mean and standard deviation over three repeats). The camera comparisons grow with speed while the reference-to-reference comparison stays flat. IMU in the figure legend denotes the ESAT gyroscope reference.

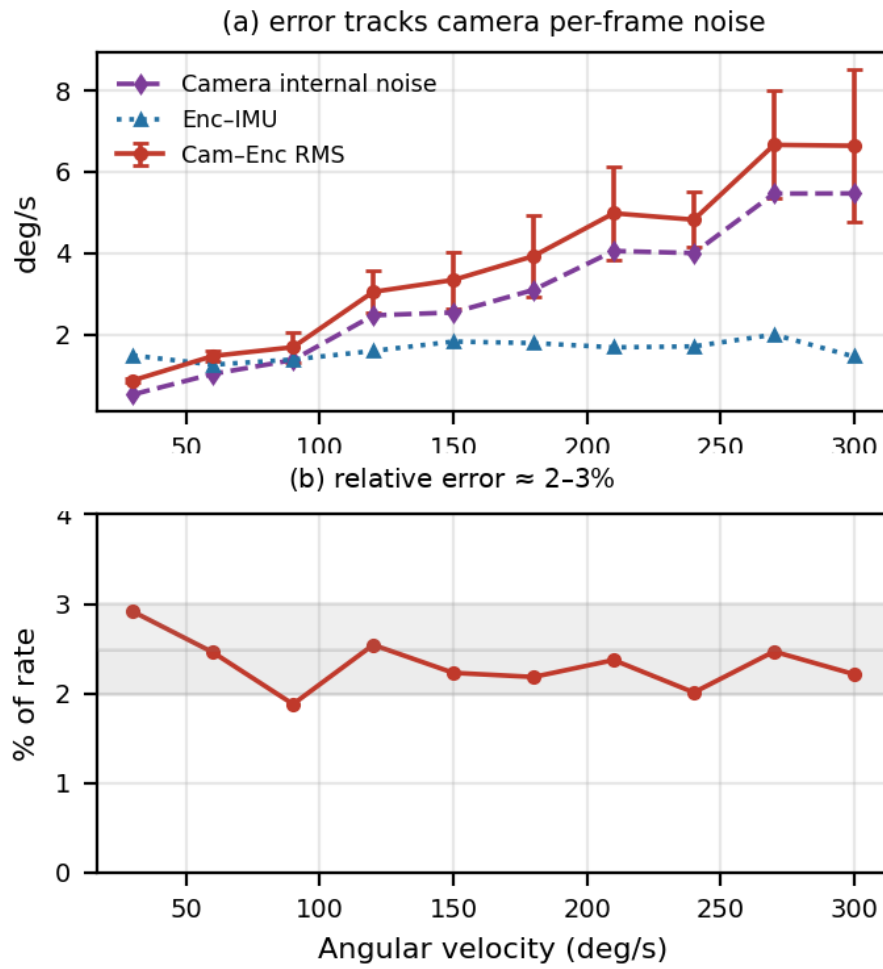


Fig. 4. (a) Camera–encoder RMS against the encoder-independent camera-side rate-variation estimate and the flat encoder–ESAT agreement. (b) Camera–encoder error as a percentage of commanded rate. Enc-IMU in the figure legend denotes the encoder-ESAT comparison.

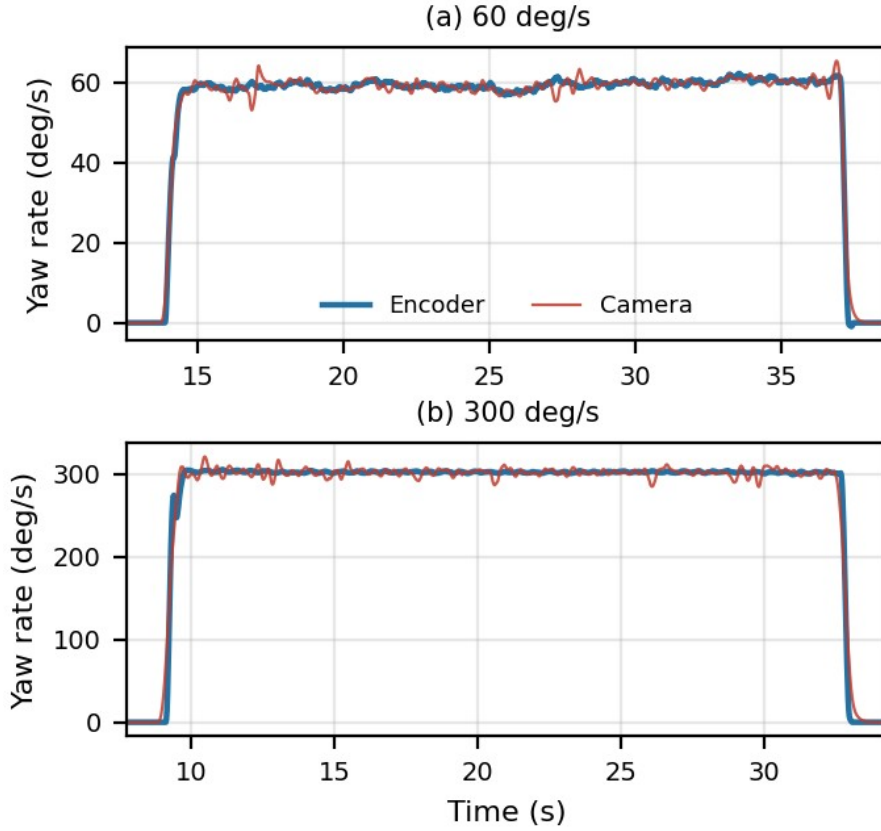


Fig. 5. Representative camera (red) and encoder (blue) yaw-rate traces at (a) 60 deg/s and (b) 300 deg/s after synchronization.

Fig. 4 separates the error into its components. The camera–encoder RMS closely follows the encoder-independent camera-side rate-variation estimate, and the error expressed as a fraction of rate is nearly constant. Representative time-series overlays at a low and a high setpoint are shown in Fig. 5.

VII. DISCUSSION

Three observations point to the same conclusion about the source of the high-rate error. First, the encoder and ESAT gyroscope rely on different physical principles, clocks, and sampling paths, and their agreement does not degrade systematically with speed. If the growing camera–encoder error were caused primarily by either reference, the encoder–ESAT comparison would be expected to grow as well; it does not. Their agreement therefore supports, but does not by itself prove, that both remain close to the platform rate.

Second, the camera–encoder RMS closely follows the encoder-independent camera-side rate-variation proxy defined by (10); the two curves rise together (Fig. 4a), which places the source of the growth inside the camera stream. Because (10) is formed from the camera yaw on the camera time base, it isolates the camera stream rather than the optics specifically.

Third, the magnitude and trend are consistent with motion blur at the fixed exposure. The angle swept by a tag edge during the exposure is $\Delta\phi = \omega t_{\text{exp}}$, which for $t_{\text{exp}} = 5$ ms gives about 0.15 deg at 30 deg/s and 1.5 deg at 300 deg/s, so the blur, and the corner-localization noise it produces, grows with rate. Fiducial-marker pose accuracy is known to degrade under motion blur [4], although that comparison applies blur synthetically to static imagery and reports pose rather than angular-rate error, so it supports the mechanism rather than its magnitude here.

Blur is not the only mechanism that predicts an error that grows with rate. Frame-to-frame timestamp jitter displaces individual

yaw samples by approximately $\omega\delta t_i$; after differentiation, timing noise with a similar statistical magnitude across runs can therefore produce rate error that increases with angular speed. The logged frame intervals give a partial view of this. Across the thirty runs the median interval is 29.9 ms with a standard deviation of 4.2 ms, 2.2 ms excluding the 0.6% of intervals that exceed one and a half times the median, and none of these statistics depends meaningfully on commanded speed. These figures characterize the host-side logged timestamp sequence; the delay between exposure and the host timestamp is not observable in this dataset, so they do not bound the true capture-time error and do not separate an optical from a timing mechanism. Doing so requires varying the exposure at fixed rate, or timestamping frames at the sensor.

Taken together, the study describes a transition. Before synchronization, inter-sensor timing dominated the apparent error. After synchronization, the remaining error is camera-stream-side and grows with rate. The evidence is consistent with camera-side acquisition effects, of which motion blur and residual frame-timestamp jitter are the two candidates the present setup does not separate. At low rates the camera agrees with the encoder to under 1 deg/s, while at higher rates the error increases and may be reduced by shortening exposure, adding illumination, increasing frame rate, or improving frame timestamping. The near-constant relative error of about 2–3% of rate is a useful empirical rule for constant-rate operation in this configuration, but it remains specific to the tag size, distance, lens, exposure, and timing implementation used here.

VIII. LIMITATIONS AND FUTURE WORK

The study validates a single axis. In our configuration the marker rotates in the image plane about the camera boresight and remains nearly fronto-parallel, which is a well-conditioned case for planar-tag pose. Pitch and roll instead tilt the marker out of plane, and

marker accuracy is known to degrade with viewing obliquity and distance [5]; the present results should therefore be read as a favorable case for the optical method rather than a claim about three-axis attitude.

Only constant-speed motion was characterized. Each run does contain a ramp, but the ramps are used to estimate the time alignment and are then trimmed from every error metric, so the dataset supports no claim about varying-rate performance. Slews and varying-rate profiles should not be expected to behave like the constant-rate case: the acquisition-stage low-pass filter introduces frequency-dependent phase delay that a single constant time shift may not fully compensate during varying-rate motion, and residual timing error also couples into the rate comparison during transients. Verifying the envelope under varying rate is the most useful single extension of this work.

The exposure was held fixed at 5 ms for all thirty runs so that image statistics and frame timing would not vary within or between runs. The consequence is that exposure was never treated as an independent variable, and the contribution of motion blur to the high-rate error is therefore not isolated. Repeating a subset of the setpoints at several exposures, with illumination adjusted to hold image brightness constant, would separate the optical contribution directly: a blur-dominated error should fall with exposure at fixed rate, whereas a timing-dominated error should not follow the same trend.

The ESAT rotational-speed reference is logged at about 1 Hz and specified at 1 deg/s accuracy [17], so it supports the encoder rather than standing as a high-resolution reference in its own right.

Finally, synchronization is performed in software by matching the motion profile, not by a hardware trigger. The re-analysis in Section IV.B shows that the choice of alignment strategy does not materially determine the steady-state plateau RMS, but the absolute timing relationship between the streams remains unmeasured, which is why no angle-domain or latency result is claimed. Recording the camera's own frame timestamps instead of host timestamps, driving the camera from an external trigger shared with the encoder logger, or logging a periodic optical timing marker would establish that relationship directly. Extending the work to pitch and roll, to varying-rate profiles, and to an exposure sweep at fixed rate is the natural next step.

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IX. CONCLUSION

We validated camera-based yaw-rate estimation on a single-axis rotational testbed against two independent references across a 30–300 deg/s envelope, with three repeats per speed and no excluded runs. After explicit synchronization, the camera agrees with a 100 Hz encoder to better than 1 deg/s at low rate and to about 2–3% of the commanded rate up to 300 deg/s, with correlation of 0.99 or higher and camera–encoder bias magnitude below 0.6 deg/s. The encoder and ESAT gyroscope agree with each other to within about 2 deg/s with no systematic dependence on speed, supporting attribution of the high-rate degradation to camera-side acquisition rather than to the references. In a paired re-analysis of all 30 runs, estimating the alignment without the constant-speed segment changed the plateau RMS by at most 0.031 deg/s. The trend is consistent with camera-side acquisition effects. With exposure fixed across all runs and frames timestamped on the host, motion blur and residual frame-timestamp jitter both remain viable explanations, and this dataset does not separate them. For anyone using a camera as a reference on an attitude testbed, temporal synchronization deserves as much attention as the optics; in our case it was the difference between an apparent 5–8 deg/s error and a real sub-1 deg/s one at the low end of the tested range. For the tested configuration and constant-rate motion, the camera provides sub-1 deg/s agreement at 30 deg/s and an approximately 2–3% error envelope over the tested range. Whether the same envelope holds during rate changes is not established here.

Data and Code Availability—The project repository is available at <https://github.com/mohammedyosri2002/yaw-rate-validation> and contains the camera-calibration script, AprilTag multi-marker fusion and acquisition script, encoder logger firmware, printable calibration and tag targets, recovered intrinsics, the raw per-run camera, encoder, and ESAT logs for all thirty runs, the offline alignment-sensitivity analysis used in Section IV.B, and the result tables.

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